

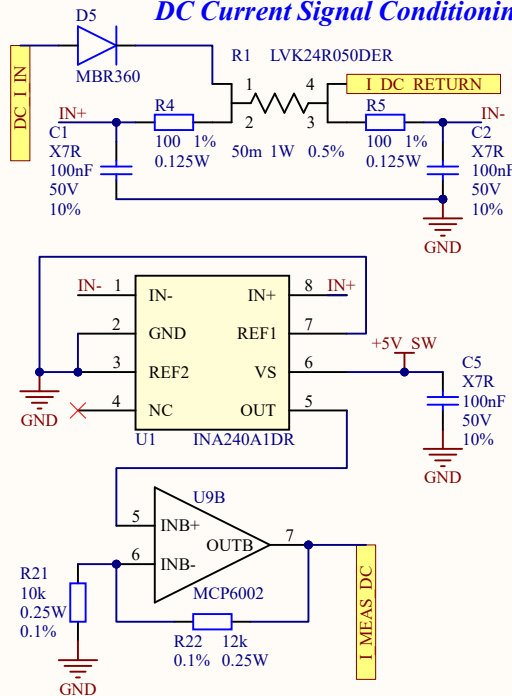
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DC Current Signal Conditioning



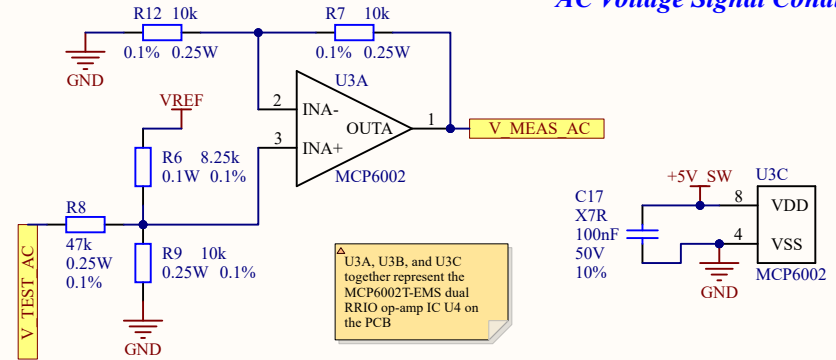
DC current input is passed through a reverse voltage protection diode in series with a 50mOhm kelin shunt resistor. Main current path returns directly to input ground via a thick dedicated trace, avoiding instability on system GND plane from high current return paths.

INA240 is a dedicated current-sense amplifier IC. + and - sense taps are difference amplified, passed into a non-inverting gain 1.2 amplifier to expand FSR, and sent to MCU ADC

AC Voltage Signal Conditioning

AC voltage input is passed through a 0.1754 voltage divider. 2.5V DC offset VREF is introduced through impedance matched R6. R6 and R8/R9 network forms a new 1/2 voltage divider, which needs to be counteracted. Signal is passed into a gain 2 non-inverting op-amp to buffer and amplify back to expected range.

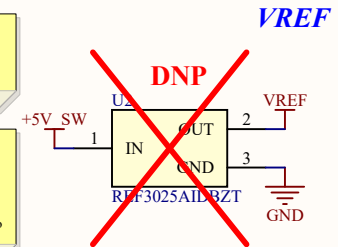
0.1754 voltage divider was chosen to work optimally with 10Vrms AC voltage input. After passing through the divider, referencing the 2.5V DC offset, and buffering/amplification, the peak to peak of the AC voltage input signal is compressed to approximately 0.02V -> 4.98V.



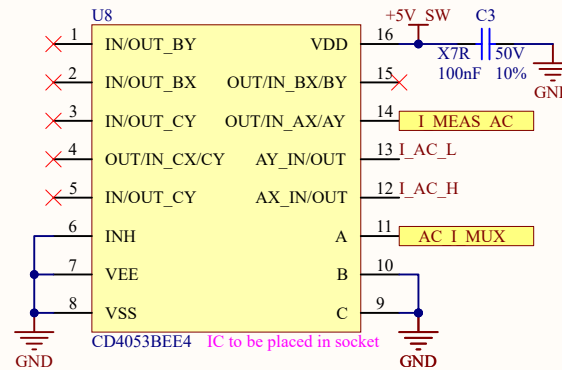
U3A, U3B, and U3C together represent the MCP6002T-EMS dual RRIO op-amp IC U4 on the PCB

Voltage from 5V power rail is passed through a 1/2 voltage divider. Halved signal is buffered through an op-amp and distributed to VREF net.

REF3025 was part of an alternate architecture, taking the 5V rail as an input and outputting a 2.5V reference voltage. In testing, the output was unstable, highly likely due to incorrect implementation. Buffered divider was found to work well, so that solution was implemented instead



AC Current Signal Conditioning

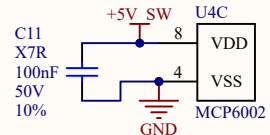


VREF introduces 2.5V DC offset to input AC signal through R16

Filtered and DC offset I_AC_IN signal is passed into non-inverting inputs of two non-inverting amplifier op-amps. One op-amp is with gain 17.5, the other with gain 1.75. Op-amp inverting inputs reference amplification to 2.5V DC offset maintaining signal center at 2.5V

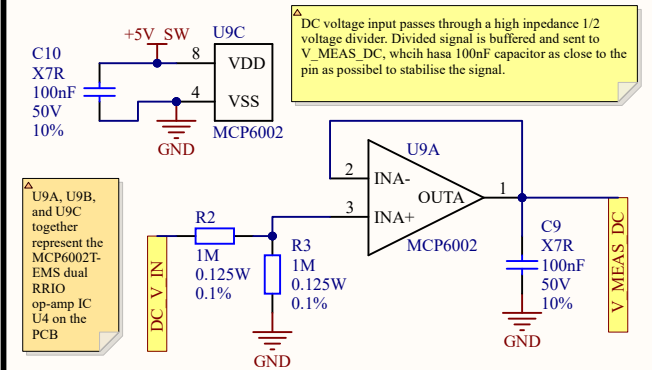
high gain op-amp output is for low current signal inputs. Low gain op-amp is for high current signal inputs. Both output signals are passed into an analog multiplexer.

U4A, U4B, and U4C together represent the MCP6002T-EMS dual RRIO op-amp IC U4 on the PCB



Multiplexer and firmware through AC_I_MUX interpret I_AC_L and I_AC_H, determining which signal to send to I_MEAS_AC, depending on high or low current state

DC Voltage Signal Conditioning



DC voltage input passes through a high impedance 1/2 voltage divider. Divided signal is buffered and sent to V_MEAS_DC, which has a 100nF capacitor as close to the pin as possible to stabilise the signal.

U9A, U9B, and U9C together represent the MCP6002T-EMS dual RRIO op-amp IC U4 on the PCB

Title			ENG2800 Team07 - Measurement Hardware
Size	Number	Revision	
A4		04	
Date:	6/20/2026	Sheet of	
File:	C:\Users\...\ENG2800 Team07 01.SchDoc	Drawn By: Kai Staby	

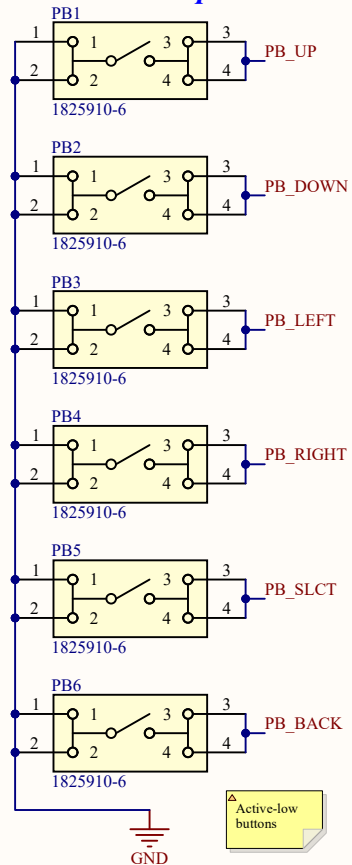
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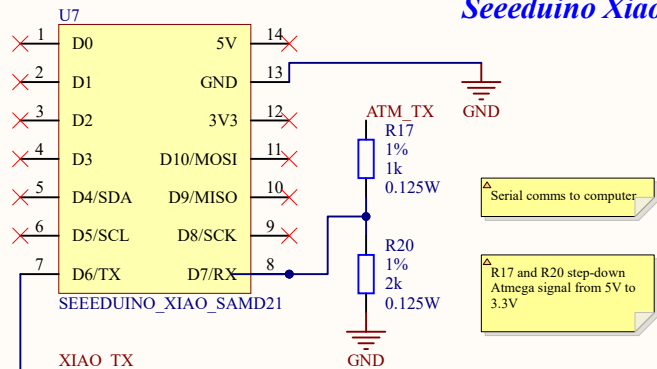
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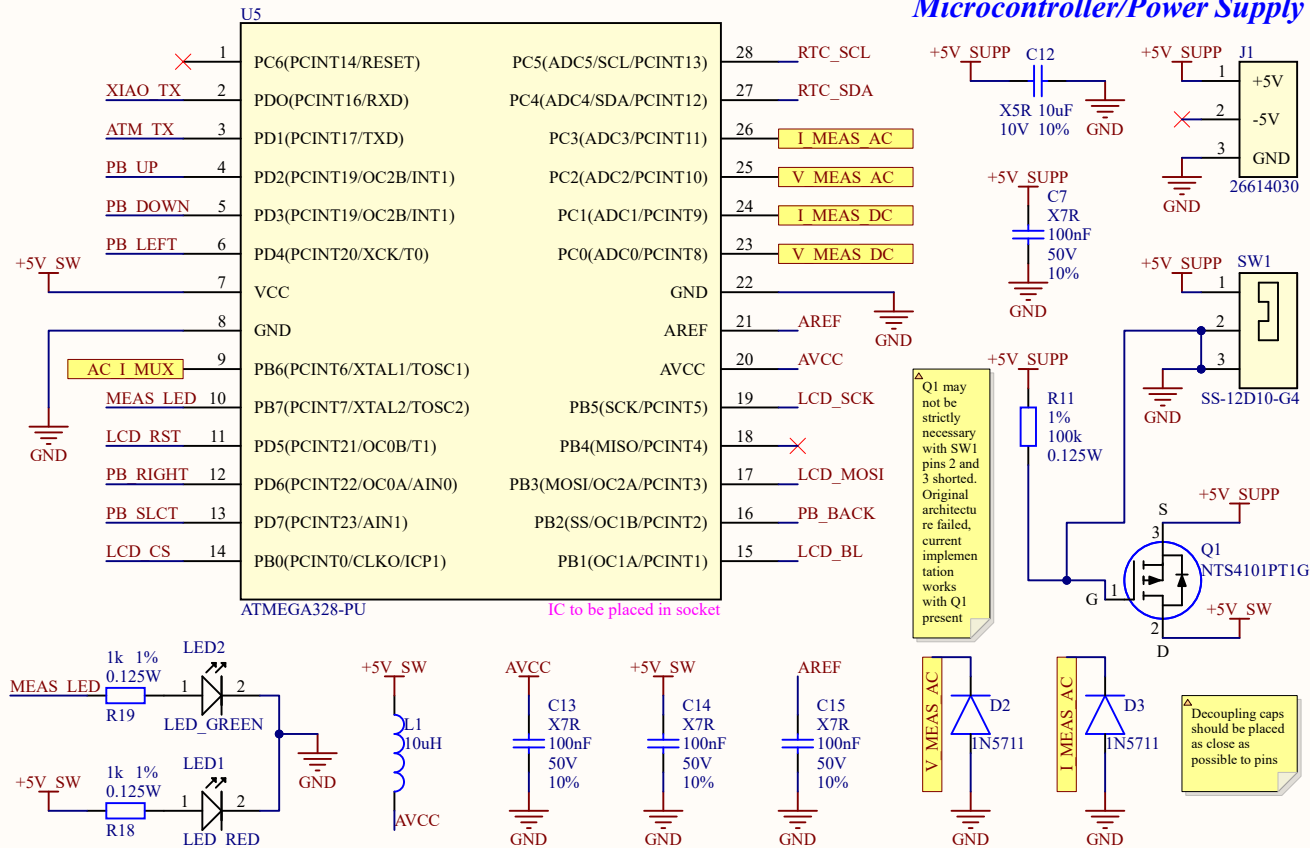
Input Buttons



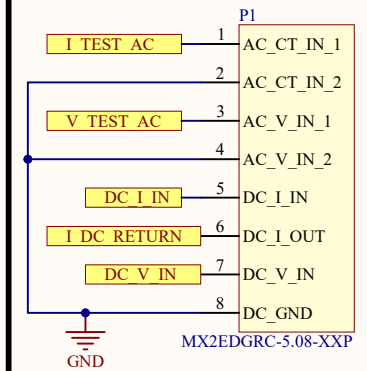
Seeeduino Xiao



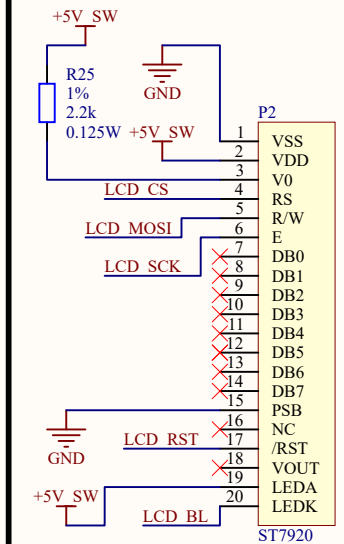
Microcontroller/Power Supply



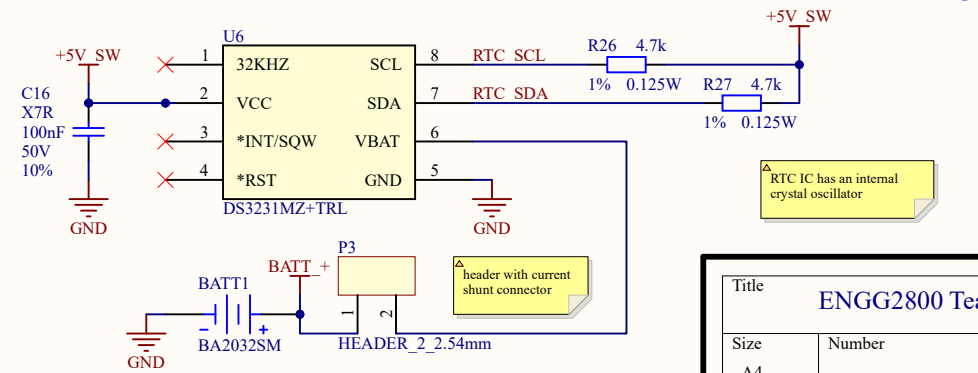
Test Harness Header



LCD



RTC



Title		
ENGG2800 Team07 - Microcontroller/Peripherals		
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Date:	6/20/2026	Sheet of
File:	C:\Users\...ENGG2800 Team07 02.Schd	Drawn By: Kai Staby

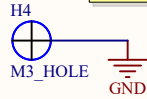
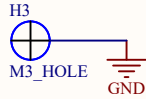
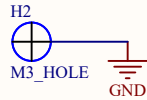
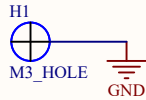
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Mounting Holes



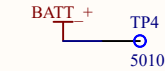
Δ Drilled into board corners

Fiducial Marks

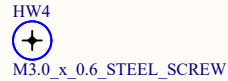
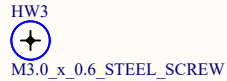


Δ Placed next to M3 corner holes, 3 per side

Test Points



Mounting Hardware



Δ Threaded from top of LCD into M3x12 brass spacer



M3.0_x_6.3_NYLON_FEMALE-FEMALE_HEX_SPACER



M3.0_x_6.3_NYLON_FEMALE-FEMALE_HEX_SPACER

Δ threaded onto nylon male-female spacer on top of board, fastening spacer in place



M3.0_x_6.3_NYLON_FEMALE-FEMALE_HEX_SPACER



M3.0_x_6.3_NYLON_FEMALE-FEMALE_HEX_SPACER



Δ Positioned on opposite side of board to M3x12 brass spacer



M3.0_x_20_NYLON_MALE-FEMALE_HEX_SPACER



M3.0_x_20_NYLON_MALE-FEMALE_HEX_SPACER

Δ Spacer mounted on the bottom of the board, threading through M3 corner holes to top of board



M3.0_x_20_NYLON_MALE-FEMALE_HEX_SPACER



M3.0_x_20_NYLON_MALE-FEMALE_HEX_SPACER



M3.0_x_12_BRASS_FEMALE-FEMALE_HEX_SPACER



M3.0_x_12_BRASS_FEMALE-FEMALE_HEX_SPACER



M3.0_x_12_BRASS_FEMALE-FEMALE_HEX_SPACER



M3.0_x_12_BRASS_FEMALE-FEMALE_HEX_SPACER

Δ Standoff between LCD and top of board



M3.0_x_10_NYLON_PAN_HEAD_SCREW



M3.0_x_10_NYLON_PAN_HEAD_SCREW

Δ Threaded through M3 nylon hex nut from bottom of board into M3x12 brass spacer on top of board



M3.0_x_10_NYLON_PAN_HEAD_SCREW



M3.0_x_10_NYLON_PAN_HEAD_SCREW

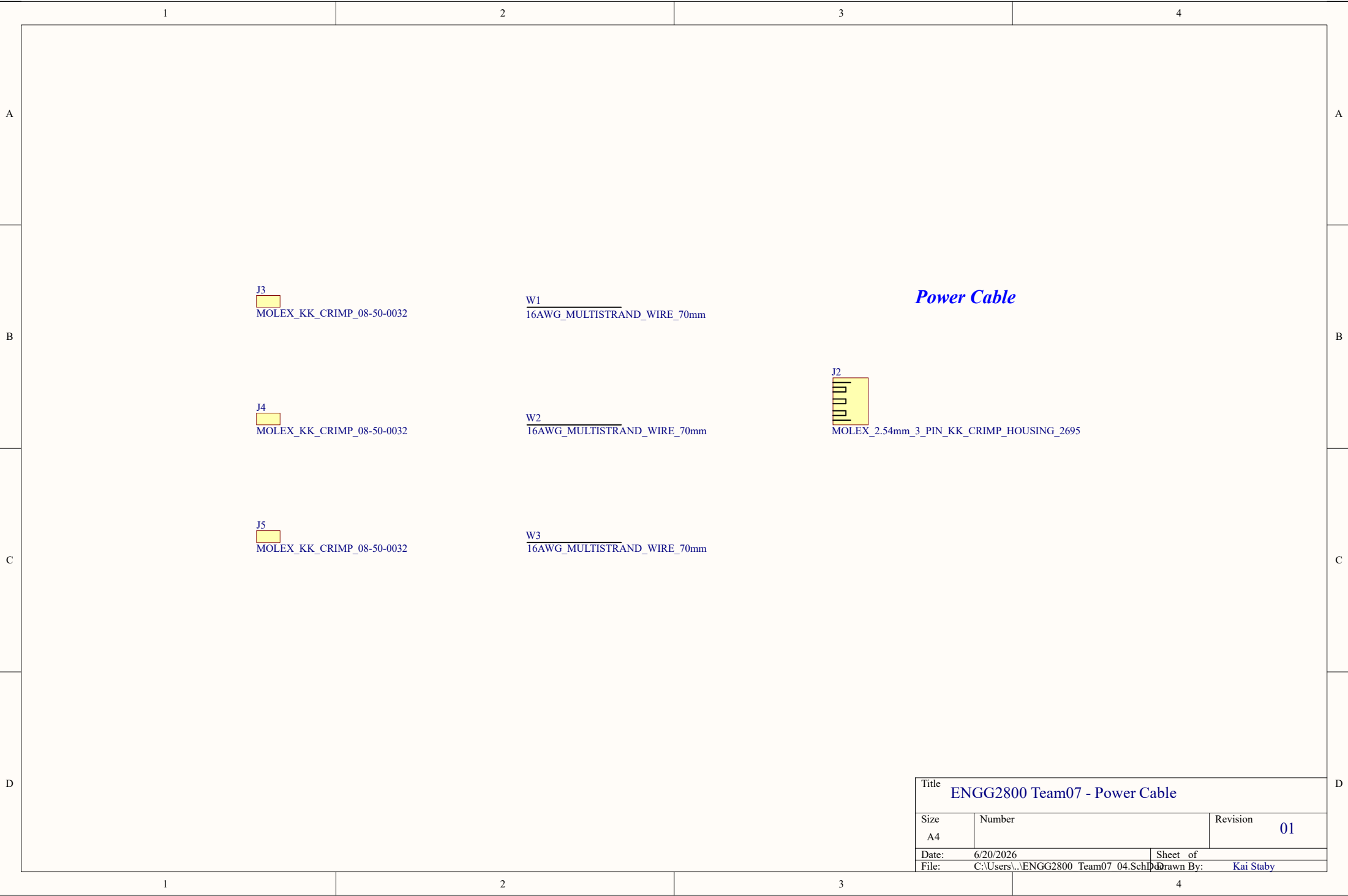
Title ENG2800 Team07 - Mounting/Test Points/Peripherals		
Size A4	Number	Revision 03
Date: 6/20/2026	Sheet of	
File: C:\Users\...\ENG2800 Team07 03.SchDoc	Drawn By: Kai Staby	

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Title ENG2800 Team07 - Power Cable		
Size A4	Number	Revision 01
Date:	6/20/2026	Sheet of
File:	C:\Users\...\ENG2800 Team07 04.SchD	Drawn By: Kai Staby